An Introduction to Flexure Design

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Abstract / Introduction

Flexures have become more and more common in aerospace mechanisms. I was first introduced to them shortly after I started working at JPL in January of 1992 and have continued to use and advocate for them ever since. While flexures have seen a continued increase in usage, many engineers, while having heard of them, don't truly understand them, or how to go about designing them. This paper is intended to introduce the novice flexure designer to the many uses, design considerations, and fabrication options for flexures along with presenting many example designs to inspire creativity and innovation in budding flexure designers.

What are Flexures?

Flexures are best described as "compliant structure". While they share many of the same attributes as springs, they differ in that flexures are part of the primary structure of an object or device. Flexures are often integral to a device whereas springs are mechanically captured in place. While springs may be in the load path, they share that load path with additional structure. A simple example which illustrates the difference between springs and flexures is the Jeep (the iconic American 4-wheel drive vehicle) suspension. Older Jeeps use leaf springs while newer Jeeps have coil springs. They both have solid, straight axles with springs to cushion the ride, but the new Jeeps with coils have trailing arms and track bars (4-bar linkages) to locate and attach the axle to the frame. If the springs were removed from this arrangement, the axle would still be connected to the vehicle and it could probably still be driven, though it may not be a pleasant ride. The older Jeep uses leaf springs to provide the suspension as well as attach and position the axle to the vehicle. The leaf springs are primary structure in this case, and if they are removed, then the axle is no longer attached to the Jeep.



Figure 1. My grandfather's vintage sheep shears. This was made from a single sheet of steel which was cut, bent, and folded as needed to produce the blades, handles, and flexure.

Metallic flexures have been around for hundreds of years, likely since shortly after people started forging steel and discovered its flexible properties. Two examples over 200 years old that immediately come to mind are the leaf spring type suspension for the bench seat on a horse drawn wagon and the hand-operated sheep shear (Figure 1).

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It wasn't until around the late 1950s or early 1960s however that flexures started to see use in aerospace structures and mechanisms. Since then, they have gained popularity and have continued to see more and more usage in the aerospace industry. This is at least partially due to more recent enabling technology such as computer-aided design (CAD) and finite element analysis (FEA) along with improved manufacturing methods and processes, like the development of Electrical Discharge Machining (EDM) techniques. [1]

While flexures can certainly be made by conventional manufacturing techniques, and a great many have been, the development of the wire EDM process drastically changed the world of flexure design. Its small kerf, high precision, and ability to cut thin blades allows for very small, intricate, and integrated flexure designs (where the flexure, the mounting structure, and the moving component are all implemented as a single part) that are simply not possible with conventional machining. The compact brake mechanism shown in Figure 2 is a good example of a device that could not be reasonably made through a process other than wire EDM. The evolving additive manufacturing (3D printing) technology promises to advance the possibilities of flexure design even further. Metallic 3D printing allows the design of overlapping and interweaving flexure elements and even changing material throughout the length of the flexure blade to tailor the design properties.

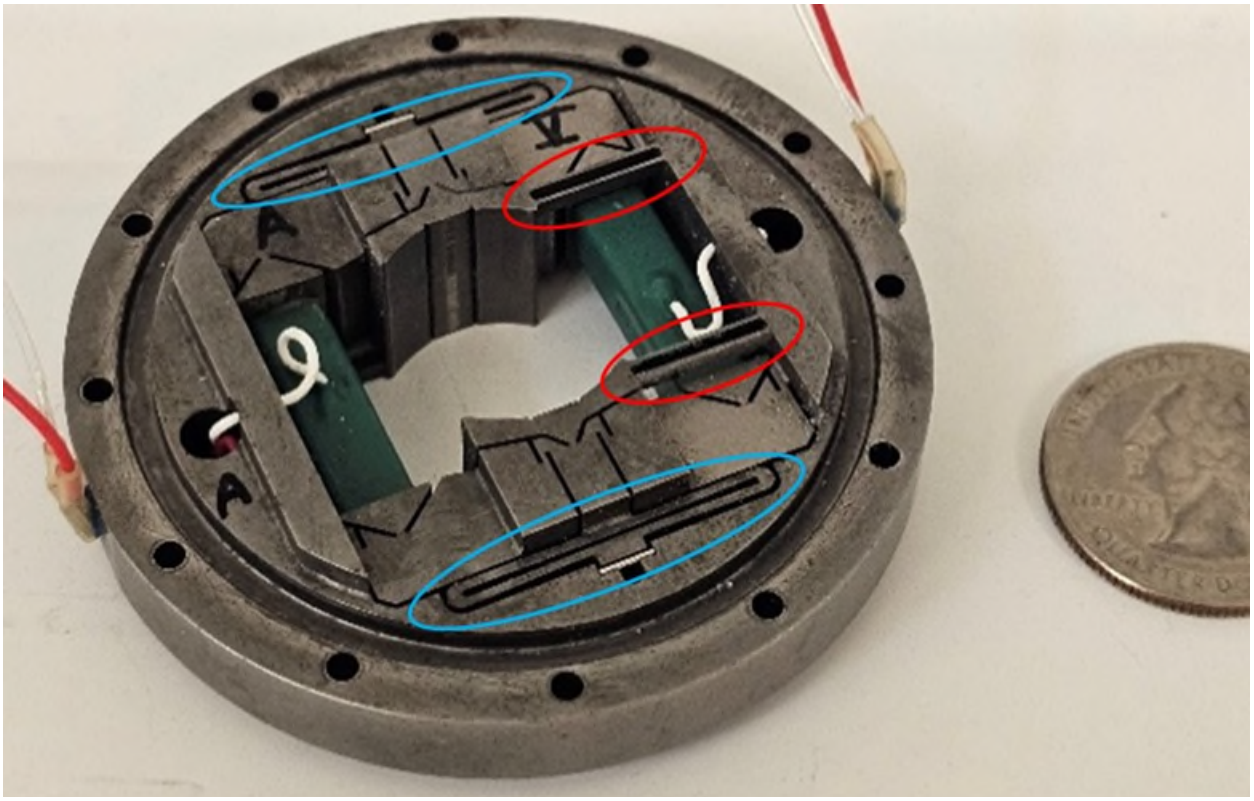


Figure 2. A piezo electric transducer (PZT) actuated brake mechanism for a coaxial rotor (not shown). Note the out of plane "tangent bars" (circled in red) which stabilize the brake shoes. The many short flexures here function like pivots. The large "racetrack" flexure blades (circled in blue) function as integral springs to preload the brake shoes against the rotor. Note the light-colored shim material inserted outboard of these springs to generate the preload.

Why flexures?

Flexures offer some very attractive features for the hardware designer. Flexures can be incorporated in mechanisms with moving joints as pivots and sliders, enabling rotation and translation without wearing parts or stiction. They are also quite useful in providing semi-kinematic mounts and allowing for relatively large

CTE mismatches or thermal gradients between parts. Because of their low cross-sectional area and long conductive path, flexures also provide good thermal isolation while still maintaining robust structural support. Along with other design elements, they can even be used to provide highly repeatable separation joints in precision devices. They are frictionless, require no lubrication, and exhibit little to no hysteresis. They can even be designed to provide infinite life with no wear or loss over time.

Flexures are typically used for linear or rotary motion, where a bending beam is employed. While less common, torsional flexures can also be designed and used. Flexure motions can be very small, highly precise, and repeatable (sub nanometer), or quite large while still providing smooth, predictable motion. Flexures are commonly coupled with high precision actuators, like PZTs, to provide preload, precise guided motion, and small output strokes. These arrangements are commonly found in opto-mechanical devices and typically range in stroke from sub nanometer to tens of microns. When coupled with actuators like voice coils or lead screws, flexure motion can range from a few microns to several millimeters, and even larger for bigger devices. Of course, these large motions require large flexures and the volume to accommodate them.

Fundamental Flexure Types

There are four different fundamental types of flexures, as illustrated in Figure 3. They are the wire, toroidal, blade, and notch (though they may be referred to by other names as well). It is important to comprehend the different degrees of freedom (DoF) that each of these fundamental flexure types constrain and why one type may be better suited to a specific application than another. Once these basics are fully understood, these fundamental types can be arranged and combined to design a flexure system (typically two or more in parallel, series, or a hybrid of both parallel and serial flexures) that meets the application's needs.

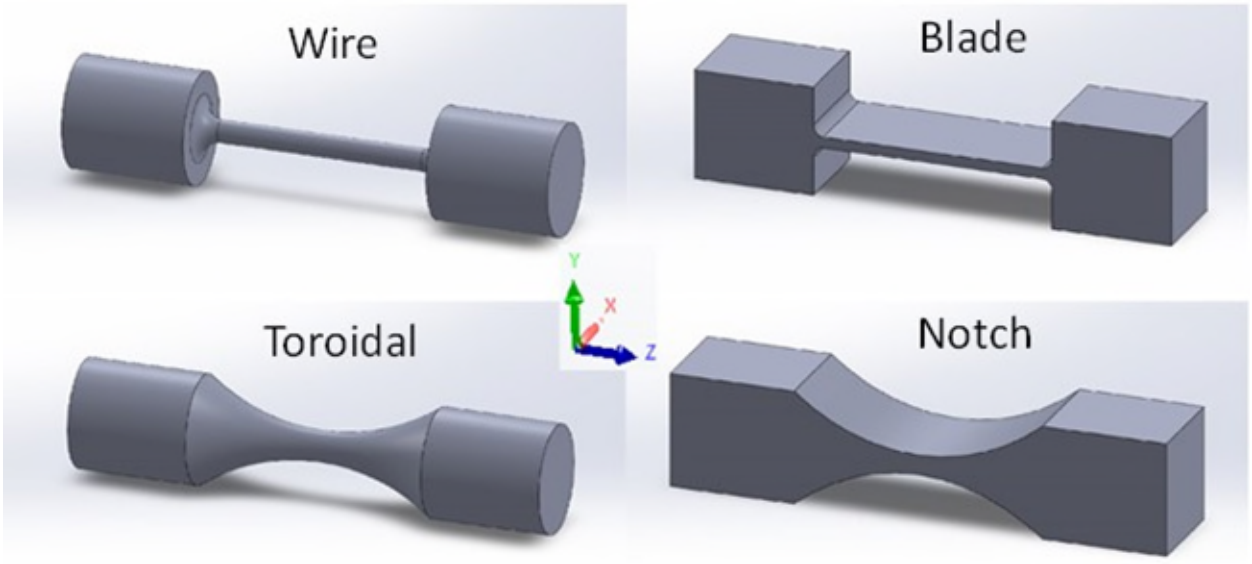


Figure 3. The Four Fundamental Flexure Types

The wire flexure only constrains 1 DoF, translation along its length (the Z axis in the figure). It is compliant in the other 5 DoFs, lateral translation (along the X and Y axes) and bending about all three axes. Because of the long, spindly nature of this type of flexure blade, it is typically loaded primarily in tension as its buckling strength is relatively low.

The toroidal flexure is like the Wire type, but because of its geometry, it can take much higher compressive loads than a similarly sized Wire flexure. The toroidal also constrains an additional two degrees of freedom for a total of 3 DoF, translation along all three axes. The toroidal flexure remains complaint in bending about

all three axes, just like the wire flexure. A common usage for the toroidal type of flexure is at the ends of struts or linear actuators where it is undesirable to translate moment loads. Sometimes both the wire and toroidal flexures are referred to as “pin” flexures.

The blade flexure (sometimes referred to as a leaf flexure) is arguably the most common type and is usually what people envision when discussing flexures. The blade flexure constrains 3 DoF, translation along the X and Z axes and rotation about the Y axis. It is compliant in translation along the Y axis and bending about the X and Z axes. This type of flexure is a good choice when long stroke is required, and the compressive loading is low.

Like the toroidal flexure, the notch flexure offers increased buckling load capability over the blade flexure and constrains an additional two degrees of freedom for a total of 5 DoF, everything but rotation about the X axis. The notch flexure can often save the day when compressive loads are high.

Initial Design

When designing a flexure, several questions must be answered. Which degrees of freedom need to be constrained, and which need to be free? Is a pivot type of flexure needed or a translation type? What is the required stroke, load magnitude, direction, and stiffness? Consideration of these questions will help guide the selection of which basic type of flexure to start with and how the flexure needs to be arranged. The following example assumes this information is known and that a simple cantilevered blade flexure is needed to provide linear motion, but the process is very similar for pivots or flexures that are fixed at one end and fully guided at the other (see Figures 5 and 6 for examples of this arrangement).



Figure 4. An example of multiple flexure blades in series to provide long stroke in a compact envelope. Note the integrated hard stop/load shunt. These linkages allowed 1 mm of linear motion and were used in an over-centering mechanism to provide reliable function without the need for adjustment. A small amount of discoloration can be seen because of excessive heat generated during the wire EDM process.

When sizing a flexure blade, there are three main parameters to work with: blade thickness, length, and width. Keep in mind that the length and thickness parameters are usually interdependent as well. For example, while the stress in a blade is primarily governed by its thickness, increasing the length of the blade reduces the amount of bending required, thus reducing the stress.

With the required displacement known, typically it is best to start a flexure design by determining the amount of length available. Don’t be afraid to get creative here. Maximize the length if needed by curving the flexible element to fully occupy the available space. Flexure blades can also be stacked in series to provide more displacement in a compact volume, as shown in Figure 4.

$$K = \frac{E \cdot b \cdot t^3}{4 \cdot l^3} \quad (1)$$

Where:
K = stiffness
E = Young's modulus
b = width of the blade flexure
t = thickness of the blade flexure
l = length of the blade flexure

Once the working length of the flexure blade has been determined, pick a blade width that fits in the design volume and a reasonable blade thickness, then calculate the resulting stiffness and stress in the blade at the maximum required stroke. This can easily be done by hand at this stage, but if a CAD program is being used to design the flexure, it may be just as easy to utilize the finite element analysis (FEA) tool in the CAD software (or even a stand-alone FEA tool), if one is available. A linear analysis is sufficient for small deflections or for providing a "first look" at flexures with larger deflections, but a non-linear analysis is recommended to give more accurate stress and stroke prediction for large deflections, especially if the flexure is fixed at one end and fully guided at the other (see Figures 5 and 6). This requires the flexure blade to form an "S" shape when deformed, which results in a change in length. To accommodate this, either the flexure blade needs to stretch, or the surrounding structure needs to deform inward (toward the flexure).

The process of initially sizing a flexure requires iteration to close in on a design that meets the stroke and stiffness requirements within a reasonable stress level. Don't get discouraged if the final design is not found right away. Change either the blade length or thickness (it's best to only change one parameter at a time, at least at first) and try again. Sometimes a design will close quickly, with only a few iterations, while other times it may feel like this process will go on forever. If a combination that works cannot be found, consider another flexure design configuration, and try again.

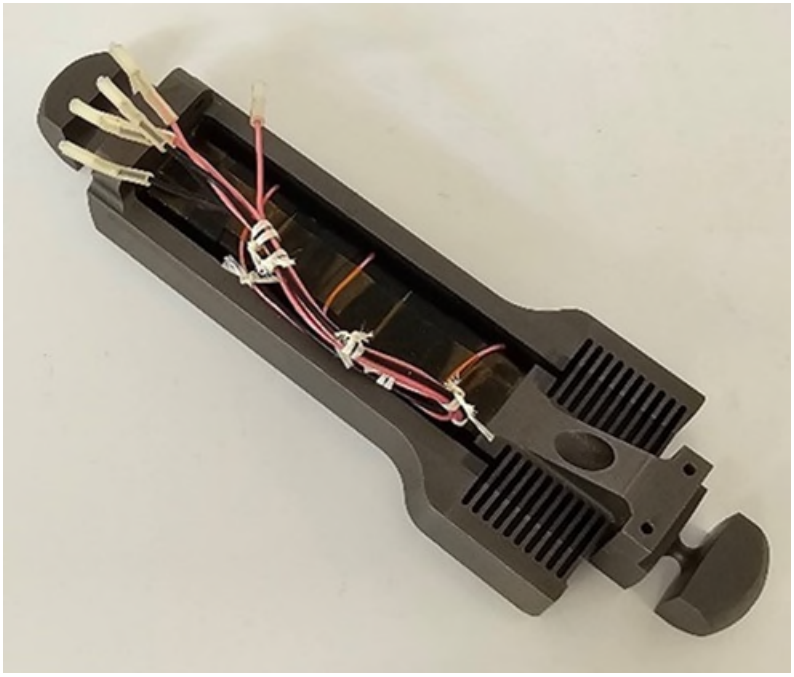


Figure 5. A PZT actuated strut with multiple parallel flexure blades. Note the toroidal flexures at the ends to reduce moment loads.

Once a flexure design is found that nearly meets the needs, the stiffness and force capability can be further refined by varying the width of the blade. Changing the width of the flexure linearly changes the load capability without impacting the stress in the flexure blade. This makes “fine tuning” a flexure relatively easy, however there are limits. For example, the width required may become too large for the packaging constraints. There are also limits in manufacturing, as the risk of deformations such as “cupping” (where the material bends or cups to form a concave surface on one side), increase as the blade width increases. (How wide a flexure blade can be machined to depends on many factors. It is best to work with a manufacturing engineer or machinist if in doubt.) If width limits are encountered, multiple blades may be arranged in parallel to provide the values needed. This approach is often employed for flexures requiring high force and relatively long stroke (where long blades are needed). See Figures 5 and 6 for examples of this parallel beam approach.

There are other design considerations to keep in mind as well. For example, flexures are sensitive to geometric tolerances. Since the thickness and length are cubic functions, the flexure stiffness is very sensitive to these dimensions. Typically, most flexures are long enough that standard tolerances are not a big concern, but they can be a factor for very short flexures. However, the blade thickness is often thin enough though that close attention must be paid to the part tolerances. Again, use of the wire EDM process is of real benefit here because of its precision. The flexure shown in Figure 6 was used in an optical instrument that required nearly perfect linear motion (no tip/tilt). The thickness tolerance of these blades was a real concern, even with wire EDM. In this case, extra blades were used to help minimize the impact of blade thickness variation. The thought was that the error would average out over the larger number of flexure blades and minimize the impact.

Another design element that requires consideration are stress concentrations at the root of the flexure, as this can be critical to cycle life. When there is a sudden change in stiffness, a highly localized stress region exists, but a generous fillet will reduce this stress concentration effect. A stress concentration factor is defined as the ratio between the maximum stress and the nominal stress. The book “Peterson’s Stress Concentration Factors” [2] goes into detail on how to determine and utilize the stress concentration factor, however it’s good practice to make these fillets as large as reasonable. A larger fillet radius will only help reduce the stress at the root of the blade. There are limitations here though, as the filet should not be made so large as to significantly impact the active length of the flexure blade.

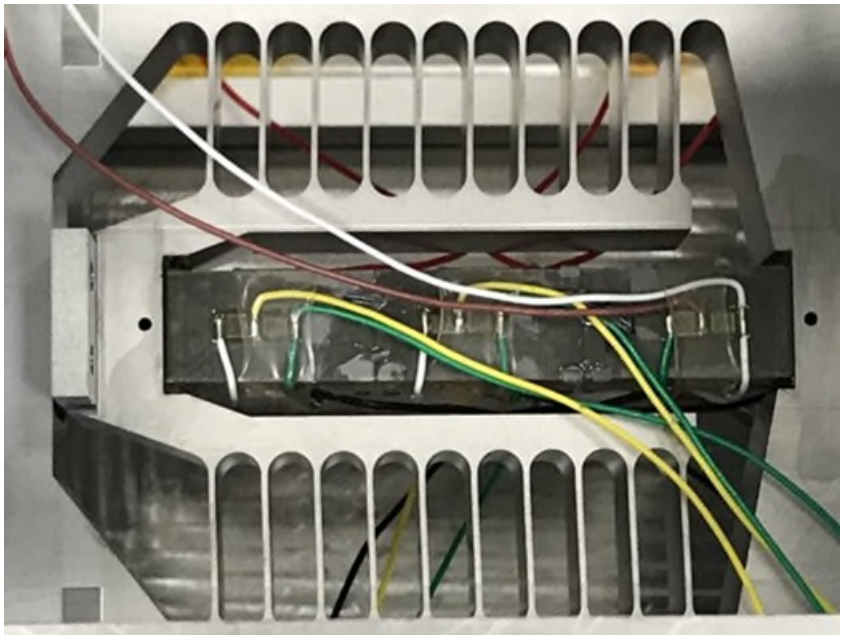


Figure 6. An example of multiple parallel blades used in an optical instrument to ensure precise linear motion.

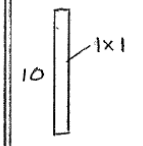
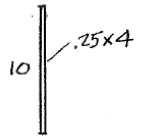
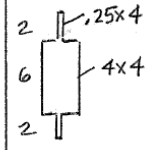
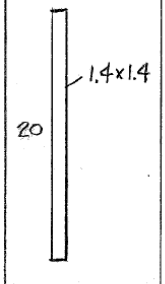
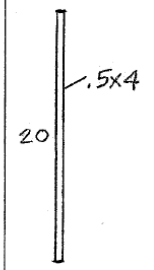
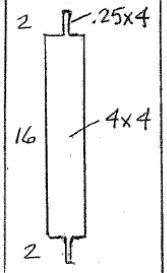
For flexures that will be loaded in compression, buckling is a critical concern. When calculating the critical buckling load for flexures, there are two important considerations to keep in mind. First, if the flexure is going through large deformations, then nonlinear buckling analysis needs to be performed. Second, if the flexure is deformed due to installation, misalignment, or functional requirements prior to the external load being applied, then the buckling analysis needs to include the deformed shape, as the critical buckling load will be drastically reduced.

If buckling is an issue, there are a few tricks that can help. One approach is to simply make the blade longer. While this may be counterintuitive, the longer flexure blade allows it to be made thicker, which helps reduce buckling. This trade doesn't always work, but it's worth trying. Another option that usually works well is to design the flexure such that the bending elements are only at the ends of the beam with a long solid section in the middle. Since most of the bending occurs at the ends of a deflecting beam, use this to your advantage. The table in Figure 7 compares these different approaches and their relative differences in stiffness and buckling strength.

Like in any good structure design, the load lines in a flexure system are important and must be managed. When designing flexures, it is critical that the load line runs down the center of the flexure blade to avoid applying moment loads. If not, the flexure will bend and move in unexpected and unwanted ways and will also be much more likely to buckle. This is especially important when designing flexure bipods for semi-kinematic mounts where they are the primary structure and under axial load. If the load line is eccentric to the flexures, the moment that is generated will significantly reduce the buckling capability of the flexures.

Fatigue is also a critical design consideration for flexures. As a rule of thumb, flexures should be designed to stay below the material's endurance limit to achieve infinite life. For spaceflight applications, NASA-STD-5001 requires a minimum service life factor of four be applied to the cycle life for fatigue assessments. This can generally be achieved by limiting the flexure's alternating stress to stay below 20% of the material yield strength. Many factors can affect the material's endurance limit though, so it's best to test the flexure design. These factors, known as a set of Marin factors, are clearly described in Shigley's "Mechanical Engineering Design" [3] and should be well understood when testing is not practical. When a flexure is subjected to random vibration excitations, the Palmgren-Miner linear damage rule, along with a fatigue life (S-N) curve, is often used to estimate the total fatigue damage for the applied stress levels and cycles. The details of how Miner's rule works can be found in "Metal Fatigue Analysis Handbook" by Yung-Li Lee, Hong-Tae Kang [4].

Flexure Comparison

Dimensions						
Strength	1	1	1	2	2	1
Axial Stiffness	1	1	2.29	1	1	2.00
Lateral Stiffness	1	.063	.080	.50	.063	.016
Axial Stiffness / Lateral Stiffness	1	16	29	2	16	125
Buckling Strength	1	.063	1.56	1	.125	1.56
Lateral Deflection for Given Stress	1	4	3.14	2.83	8	7.80
Energy Storage	1	1	.79	4	4	.97

D. Moore 5/8/2004

Figure 7. Flexure Comparison Table generated by Don Moore, an expert in flexible structures and opto-mechanical design at NASA's Jet Propulsion Laboratory. (The dimensionless numbers in this table are for relative comparisons between the flexure options shown.

Fabrication

When designing flexures, it is important to keep in mind how they will be fabricated as this often will influence many parameters of the overall design. Flexures can be made by many different manufacturing processes. Some common methods are milling, laser cutting, abrasive water jet cutting, etching, wire and plunge EDM, and cutting or stamping blades then bonding, clamping, or welding them in place. Each of these various methods has different design restrictions like material type, blade thickness, cutting depth, and spacing between blades, just to name a few.

Flexures used in aerospace and precision mechanisms are often made using the wire EDM process. The development of this type of machining was a complete game changer for flexures and opened a whole new world of design opportunities. For very small or densely packed flexures, wire EDM is often the only practical way to go. Wire EDM machines can reliably cut most materials using a 0.15 mm (.006 in) diameter wire (which yields a kerf of about 0.20 mm to 0.25 mm (.008 in to .010 in) and hold tolerances of ±0.18 mm (±.0005 in). (Wires as small as 20 µm (.0008 in) in diameter can be used and tolerances as small as ±5 µm (+/-.0002 in) can be held under certain circumstances using the latest machines.) This small kerf and ability to hold tight tolerances allows for very intricate and compact designs. The EDM process also allows production of very thin blades because, unlike conventional milling, no significant side forces are exerted on the material during the cutting process. Just how thin a blade can be cut is dependent on many factors, but generally a flexure width to thickness ratio of 50:1 can be easily cut in most common flexure materials with the wire EDM process. This is a good starting point but be aware that this thickness to width ratio can

be pushed much further under ideal conditions, as shown in Figure 8. Work closely with a machinist or manufacturing engineer to determine what may be possible for a particular application.

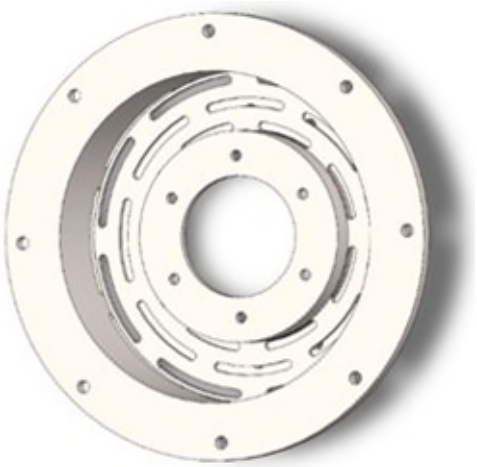


Figure 8. A simple diaphragm flexure designed to be machined conventionally on a lathe and a mill.



Figure 9. My very first flexure design. Versions of this diaphragm flexure design were cut by wire EDM and by milling.

While the wire does offer a lot of advantages for machining flexures, there are a few limitations. This process does require full access to both sides of the cut because the wire must pass through the material in a straight line (similar to a band saw blade). Because the cut is made using an electric arc, the EDM process can only be used on electrically conductive materials. Also, while the EDM machine can cut the hardened condition of metals just as easily as the annealed condition, the temperature of the electric arc is 10,000 degrees C. This heat is managed by flowing de-ionized water around the wire and through the cut, but it can locally affect the temper of the material. This is called the heat affected zone (often referred to as the HAZ) and it can extend as much as 0.38 mm (.015 in) into the material when using older EDM machines. When using modern wire EDM machines however, the HAZ is significantly reduced, approaching a near zero depth under ideal conditions.

If there is only access to one side of the part being machined, the plunge EDM process may be utilized. An electrode must first be machined, which is then used to burn the part. This electrode is slowly eroded through use, so the cut dimensions can change during the manufacturing process. This often requires the use of multiple electrodes and cutting passes. The plunge EDM process offers many of the same advantages as wire EDM, but it is used less frequently than the wire due to the added cost and time of making the electrodes. Still, this process can be very helpful when machining access is limited.

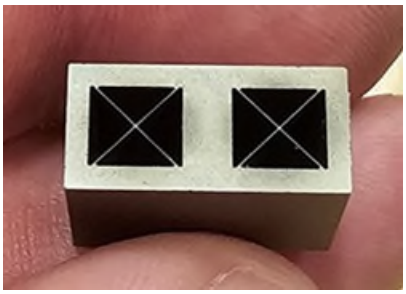


Figure 10. The blades in this steel block are 12.7-mm (.5-in) wide and only 0.05-mm (.002-in) thick, a width to thickness ratio of 250:1.

Another area that needs to be considered when using EDM is the recast layer. The recast layer is a thin layer that develops on the cut surface during the EDM process as the metal is melted by the electric arc. Most of this molten metal is flushed away, but a few particles end up getting redeposited on the parent material where is fuses to the surface. The presence of this thin metal layer on the flexures tends to generate micro-cracks when the flexures are stressed. These micro-cracks induce stress concentrations and can greatly reduce the flexure fatigue life. Hence, it is essential to remove the recast layer on all flight flexures, highly stressed flexures, or any flexure system that needs to provide a high cycle life or high reliability.

The recast layer can be removed through abrasion if sufficient access is available, but it is typically removed by chemical etching. Regardless of the method used, this is something that must be planned for as it adds cost and lead time to the machining process. A thickness allowance must be added to the flexure to account for the material being removed by abrasion or chemical etching. Newer wire EDM machines and processes are capable of making cuts with very little recast layer formation. This reduces the amount of material that needs to be removed and I would even consider forgoing the recast removal process completely on non-critical flexures with low cycle life (i.e., non-flight, lab test only applications were a flexure failure would not damage critical hardware or impact schedule).



Figure 11. The Free Flex Pivot flexure from Riverhawk (originally designed by Bendix)

Besides the EDM process, many other options exist for machining and manufacturing flexures. Take for example the Free Flex Pivot flexure (Figure 11) originally developed by Bendix in the early ‘60s. This flexure consists of blades cut from flat stock and then welded to cylindrical sections to provide rotary motion. [5] Similarly, flexures can be made by cutting or even stamping blades from material such as shim stock and then clamping or even bonding them into place. If the blade geometry allows, conventional milling can be utilized as well. Plenty of flexures have been made this way. The water jet method of machining can be very attractive for flexures that don’t require high precision or fine surface finish. The relatively low cost of this process makes it a good choice for demonstration models or prototypes. The rough surface left from this process can cause issues similar to the recast layer discussed above, therefore it should not be considered for critical flexures (flight use, high stress, high cycle life, etc.).

The flexure design depicted in Figure 12 is made by clamping high strength steel to aluminum mounts using simple bolts and doubler plates. This allows the use of very high strength (2413 MPa (350 ksi)) flexure blades with a very lightweight system. In addition to the mass savings over an all-steel design, this approach is very simple to machine and assemble, resulting in a quick, low-cost solution. This is a great option for prototyping and test fixtures. One potential issue to be aware of with this design is the possibility of the blades slipping under high loading conditions.

Metallic additive manufacturing promises to open new flexure design options just like the development of wire EDM. Being able to print flexure blades allows the creation of geometry that cannot be machined in any way. Imagine interlaced flexure blades, multiple out of plane blades, blades that transition from one material to another along the length of the blade, and many other exciting possibilities!

Regardless of the type of machining used, precision flexures will often require rough cutting, then stress relieving, followed by final machining. Sometimes multiple stress relieving steps are required. Flexures will also often require some post processing, such as the previously mentioned recast removal or shot peening, which may also be used to relieve internal stresses and reduce crack propagation.

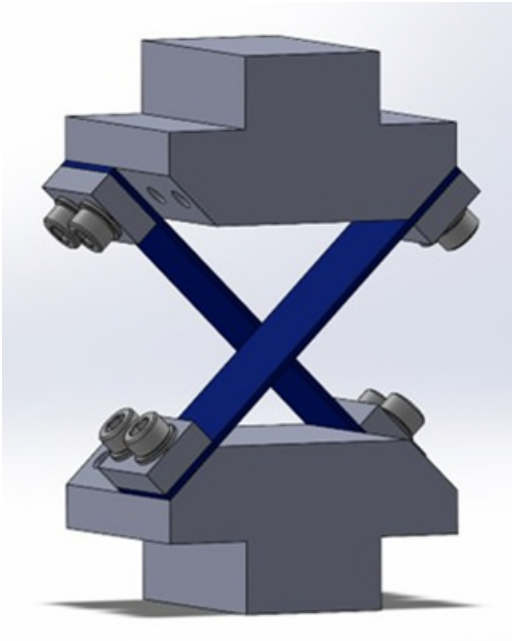


Figure 12. A Simple Cross Blade Flexure. This flexure is designed to be made by cutting hardened steel shim stock and clamping them to aluminum bases.

Materials

Flexures can be made from all sorts of materials, especially if the stress and life cycle count is low. Titanium 6Al-4V is very often used because of its light weight, high strength, and favorable endurance limit, but flexures have been made from 7075 aluminum, copper-beryllium, G-10 fiberglass, various plastics, and steels, just to name a few. In general, if the material is a good choice for a spring, it will work well for a flexure.

When selecting a material for flexures, it generally needs to be strong yet flexible, which means the material needs to have a high young's modulus to yield strength ratio. Other material properties that may need to be considered, depending on the application, are thermal diffusivity, coefficient of thermal expansion (CTE), and density when the flexures need to satisfy both thermal and dynamic environment requirements.

Center Shift

One of the disadvantages of pivot flexures is center shift. When these types of flexures are rotated, the blades are deformed in an "S" shape. Since the length of the blade does not stretch, the pivot point is forced to move away from the initial center. The amount of this shift in the center point is directly proportional to the rotation angle of the flexure.

This center shift phenomenon can be minimized by constraining the flexure blades such that they are forced to stretch. (See Figure 13 for an example.) This design approach can virtually eliminate center shift, but because the beam stretch increases the stress significantly, the stiffness is increased considerably as well, and the achievable angular displacement is much lower. Additional flexure elements can be added to reduce the blade stretch, thus increasing angular displacement, but at a cost of reduced translational stiffness. This may result in lateral motion of the center point though, especially when subjected to lateral loads.

Integral Travel Limits

When possible, it is recommended to incorporate travel limits into your flexure designs. This is especially true of small flexures with thin blades that can easily be deformed by hand. The travel limits should be sized to prevent the flexure material from yielding. Typically, these limits are set just slightly beyond the necessary travel, but sometimes in small flexures the size of the kerf is significantly larger than what is needed for the travel requirements. Also be aware that while these travel limits may help protect the flexures, they do not protect the rest of the structure. Once the travel limit is reached, the load is now reacted directly into the structure. Make sure to analyze for this and design appropriately.

Inflection Point

When flexure blades pass through their inflection point (the zero-stress position) when changing direction, a small non-linearity may be observed. This may also be referred to as “snap through”. This is typically not a problem for less sensitive devices with large ranges of motion, but it is likely to be an issue for highly sensitive mechanisms which are controlled to nanometers. Optical mechanisms that utilize flexures often restrict their motion to one direction only and limit the range of travel such that the flexure does not reach its inflection point. This also has the added benefit of eliminating stress reversal in the flexure, which will increase the cycle life.



Figure 13. A pivot flexure with minimized center shift and integral travel limits.

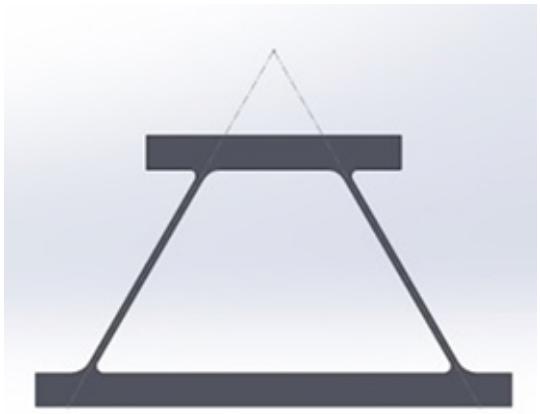


Figure 14. This Virtual Center Flexure pivots about the origin point (where the load lines intersect).

Miscellaneous Tips and Tricks

Sometimes it is desirable to add damping to a flexure. This can be achieved several ways. One method is to simply bond some viscoelastic material to the surface of the flexure blades. This approach provides some damping, but it is much more effective to utilize constrained layer damping. Returning to the leaf spring flexure example from the beginning of this paper, placing flexure blades in contact with each other and constraining them such that they are forced to move together provides damping. This approach also results in some stiction and particle generation from the blades scrubbing against each other, neither of which are desirable for precision mechanisms. However, if a layer of viscoelastic film is bonded between those same blades, good damping is obtained without the stiction or scrubbing.

A pair of blade flexures can be used to mount an object. If those blades are angled and used like a bi-pod mount, the supported object will pivot about the point where those two blades intersect. This can be used to the flexure designers benefit by making virtual pivot points where desired. Unlike when using bearings or pins, the pivot point does not need to be contained within the structure’s envelope. By using flexures, this pivot point can be projected well outside the structure. This little trick is illustrated in Figure 14. Let’s say a sensor with X thickness must be pivoted about its front surface. By attaching this sensor to the bipod flexure shown in the figure with the flexure centerlines crossing X distance from the mounting surface, this desired virtual pivot point can be achieved without having any structure protruding past the mounting surface. This sort of arrangement is still subject to the center shift phenomenon discussed earlier.

A flexure's stiffness is only as good as its base. When designing any type of flexure, pay attention to how it is secured or attached. A thick, stiff flexure blade that is attached to a soft base may not give the strength or stiffness expected. The same is true when modeling or calculating the flexure blade stress, stiffness, and stroke. Make certain that the base is accurately represented in the analysis.



Figure 15. An example of a flat spring that followed a similar design process as a flexure. This part is about 25 mm in diameter.

The item shown in Figure 15 is not truly a flexure by our definition but a flat spring. It is included here because, while this type of spring is less efficient than a common helical compression spring, it can be very advantageous for certain designs due to its packaging options. This type of spring is designed using the same process, analysis, and design considerations as a flexure.

Be cautious when designing stiff housings that contain flexure blades that are fixed at one end and fully guided at the other. As stated previously, this type of configuration requires the blades to stretch, which increases stress and reduces stroke. When this sort of flexure design is needed though, adding a small perpendicular flexure blade at the end of the main blade (as shown in Figure 16) will reduce the stress and increase the stroke with only a small reduction in the overall stiffness.

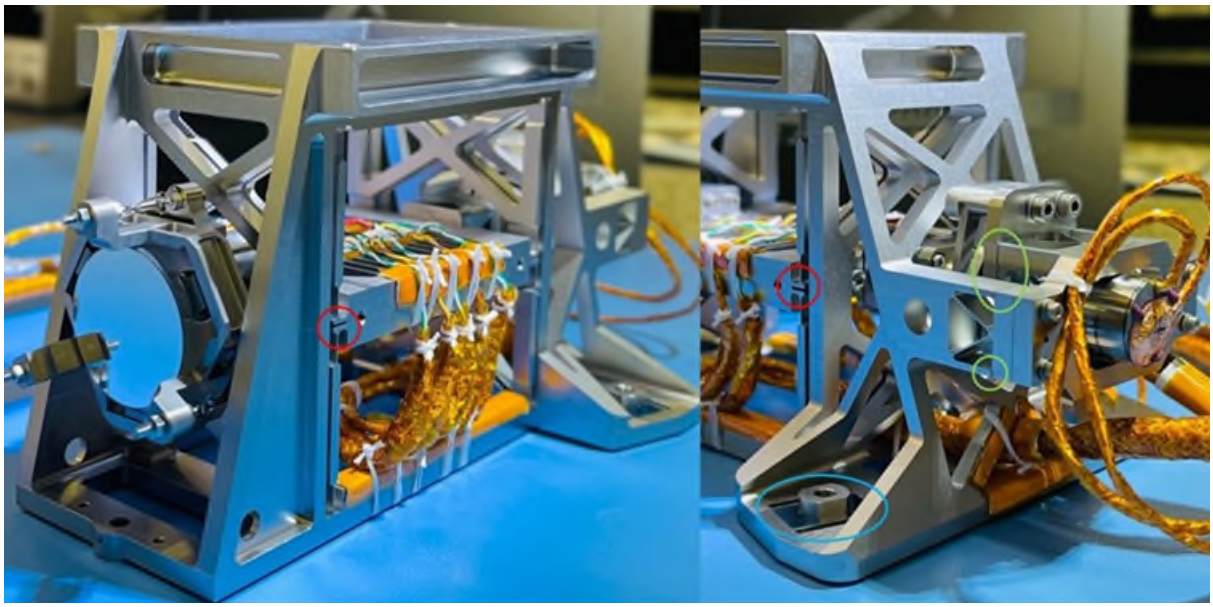


Figure 16. The Focus Control Mechanism for the Coronagraph Instrument contains many different flexure elements. Note the small flexures at the inner ends of the main flexures (circled in red). These small blades significantly reduced the stress in the main flexures and allowed a larger stroke. The small gaps circled in green serve as travel limits for the motor mount flexures. The "H" shaped flexures circled in blue allow for thermal expansion

The remaining pictures are provided to inspire ideas and possible solutions to flexure design problems. These are all flight designs developed for various missions and demonstrate some unique approaches to flexure design.



Figure 17. A flexured rod that makes up part of a precision mount assembly commonly used at JPL on opto-mechanical devices. It is referred to as a "Moore Mount" after its originator, Don Moore. These were machined using an end mill.

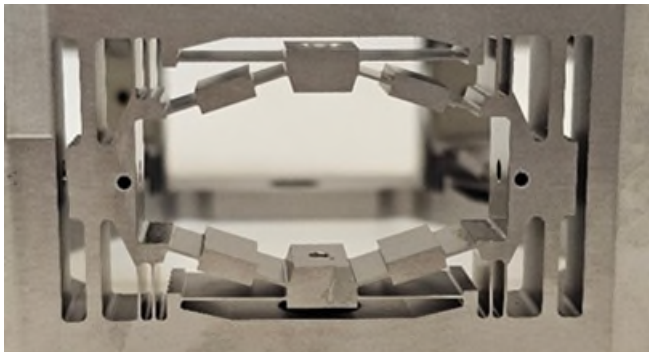


Figure 18. A "scissor jack" flexure arrangement demonstrating many of the design concepts discussed. This is part of an opto-mechanical mechanism originally designed for a space interferometer. It was wired out of a single piece of titanium. The flexure arrangement shown is approximately 50-mm wide.



Figure 19. A simple cantilevered flexure design that is "folded" to gain more length in the volume allowed. This was designed to be used for the Mars Sample Return containment vessel.



Figure 20. Three different Mars rover wheels, all using flexures to provide a simple suspension.

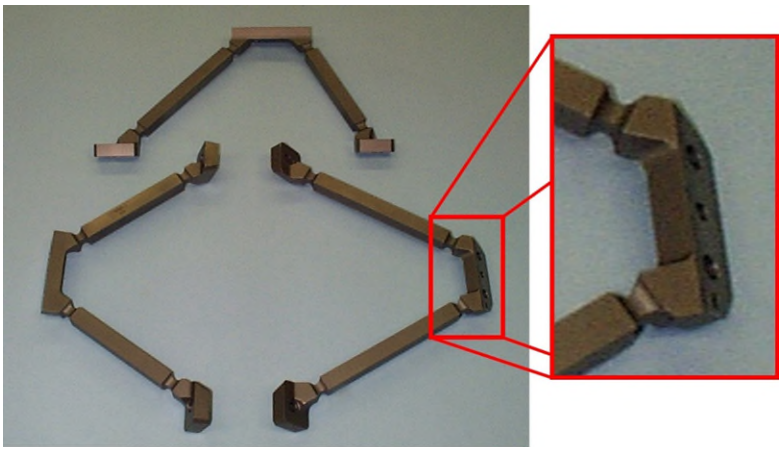


Figure 21. Flexured mounting bipods designed to mount a laser assembly to the Microwave Limb Sounder THz instrument optical bench. These were machined conventionally from Ti 6Al-4v. Note the hybrid notch/toroidal flexures.



Figure 22. A flexured rod guide. The extremely low cycle requirement allowed the use of aluminum 7075 for this part.

Conclusion

There is often a need to add compliance to modern aerospace structures and mechanisms. This is often particularly needed for opto-mechanical devices, high precision mechanisms, and even designs needing materials with different coefficients of thermal expansion and will experience a wide temperature excursion. Having a solid understanding of the fundamental flexure types and their characteristics is essential. The basic concepts and design considerations presented in this paper were meant to provide guidance during the process of developing flexure-based solutions to meet challenging design requirements, but this is just the beginning. I sincerely hope that this paper, along with the additional references listed, has inspired the reader to continue investigating the nuances and challenges of flexure design and the many variants that exist. The constant improvements in fabrication and the budding potential of additive manufacturing promise to open new possibilities and design options for the creative flexure designer.

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